

# NTRK3 fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes NTRK3 gene fusions from the msk\_impact\_50k\_2026 study, covering 58 fusion events. 55/56 fusions with known frame status (98.2%) were in-frame. 55/57 fusions (96.5%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ( $p=0.0695489$ , not statistically significant at  $\alpha=0.05$ ). No literature benchmark is configured for NTRK3.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 58 analyzed fusion events, identifying 4 distinct fusion partner genes. The most frequent partner was ETV6, observed in 55 events.

### Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 54/55 (98.2%) of in-frame fusions retained the domain,  $p=0.0695489$  (not statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

### Domain disruption

Domain-disruption analysis was skipped: NTRK3 has no disruption\_required\_domains configured; domain-disruption analysis was skipped.

### Cutpoint detection

Cutpoint detection scanned 57 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 155 aa (exon 5/6 boundary; permutation-corrected  $p=0.036963$ , statistically significant at  $\alpha=0.05$ ). This is -5 aa from the nearest configured domain boundary at 160 aa.

### Corroborating confidence statistics

Corroborating confidence statistics compared Domain\_retention\_flags groups "retained" ( $n=55$ ) and "not\_retained" ( $n=2$ ). For Frame\_status, retained: 54/54 (100.0%, 95% CI 93.4%-100.0%); not\_retained: 1/2 (50.0%, 95% CI 9.5%-90.5%). Welch's t-test comparing Tumor\_variant\_count between groups gave means of 74.1455 vs 7,  $p=5.97753e-12$  (statistically significant at  $\alpha=0.05$ ).

### Composite evidence score

Composite evidence score reported: domain\_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption\_required\_domains not configured for this gene).

### Exon retention

Exon retention produced results for this run (1 row(s) in Exon\_retention); see the accompanying table.

## **Expression association**

Expression association reported: No mRNA expression data was available for NTRK3 in this cohort; expression-association analysis was skipped.

## **Genomic position recurrence**

Genomic position recurrence produced results for this run (33 row(s) in genomic\_bin\_recurrence, 9 row(s) in exact\_position\_recurrence, 2 row(s) in protein\_position\_genomic\_spread); see the accompanying table.

## **Joint partner**

Joint partner reported: NTRK3 has no gene\_pair configured; joint-partner analysis was skipped.

## **Mechanistic interpretation**

Mechanistic interpretation produced results for this run; see the accompanying data for details.

## **Mutation cooccurrence**

Mutation cooccurrence reported: NTRK3 has no mutual\_exclusivity\_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

## **Window detection**

Window detection produced results for this run (18 row(s) in window\_scan, 4 row(s) in top\_windows); see the accompanying table.

# Tables

## composite\_score tables

composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                           | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|---------------------------------------------------------------------------------|---------------------|------|
| ETV6         | 55          | 0.9482758620689655   | 0.20043213737826426    | None                    | 0.0                      | 0.5613244116887826         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.4884040426676675  | 1    |
| DOT1L        | 1           | 0.017241379310344827 | 0.20043213737826426    | None                    | 1.0                      | 0.5613244116887826         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.3267661116331847  | 2    |
| VPS39        | 1           | 0.017241379310344827 | 0.20043213737826426    | None                    | None                     | 0.5613244116887826         | confidence_certainty,<br>domain_retention,<br>recurrence                        | 0.17140444508699657 | 3    |
| ZNF710       | 1           | 0.017241379310344827 | 0.20043213737826426    | None                    | 0.11159235668789813      | 0.5613244116887826         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.1601896785121656  | 4    |

## cutpoint\_detection tables

cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio | p_value             | neg_log10_p_value  |
|----------|------------------------|------------------|------------------------|------------------|------------|---------------------|--------------------|
| 155      | 0                      | 55               | 1                      | 1                | 0.0        | 0.03508771929824562 | 1.4548448600085102 |
| 466      | 14                     | 41               | 2                      | 0                | 0.0        | 0.07518796992481205 | 1.1238516409670856 |
| 472      | 15                     | 40               | 2                      | 0                | 0.0        | 0.08521303258145364 | 1.069493978644493  |

## domain\_retention tables

frame\_domain\_contingency\_table

|    |   |
|----|---|
| 54 | 1 |
| 1  | 1 |

domain\_retention\_descriptives

| Domain_key | Domain_name           | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|------------|-----------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| kinase     | Protein kinase domain | 57                      | 55                   | 0               | 2                | 0.0                                             |

## exon\_retention tables

Exon\_retention

| Exon | Retained_event_count | Total_event_count | Retained_fraction  |
|------|----------------------|-------------------|--------------------|
| 14   | 55                   | 58                | 0.9482758620689655 |

## frequency tables

Partner\_gene\_counts

| Partner_gene | Event_count |
|--------------|-------------|
| DOT1L        | 1           |
| ETV6         | 55          |
| VPS39        | 1           |
| ZNF710       | 1           |

## genomic\_position\_recurrence tables

genomic\_bin\_recurrence

| chromosome | bin_start | bin_end  | n_events | event_ids                                                                                                                                                   |
|------------|-----------|----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15         | 88506000  | 88507000 | 6        | EVT-P-0002978-T02-IM5-43,<br>EVT-P-0002978-T03-IM5-45,<br>EVT-P-0002978-T04-IM5-55,<br>EVT-P-0002978-T05-IM6-56,<br>EVT-P-0002978-T06-IM6-21, ... (6 total) |
| 15         | 88505000  | 88506000 | 5        | EVT-P-0002671-T01-IM3-30,<br>EVT-P-0035767-T01-IM6-52,<br>EVT-P-0035767-T04-IM6-10,<br>EVT-P-0035767-T06-IM7-9,<br>EVT-P-0055703-T01-IM6-5                  |
| 15         | 88488000  | 88489000 | 4        | EVT-P-0016507-T01-IM6-26,<br>EVT-P-0016507-T02-IM6-27,<br>EVT-P-0016507-T03-IM6-28,<br>EVT-P-0016507-T04-IM6-29                                             |
| 15         | 88516000  | 88517000 | 4        | EVT-P-0033921-T02-IM6-13,<br>EVT-P-0033921-T03-IM6-48,<br>EVT-P-0033921-T04-IM6-12,<br>EVT-P-0033921-T05-IM7-11                                             |
| 15         | 88501000  | 88502000 | 3        | EVT-P-0013196-T02-IM6-33,<br>EVT-P-0013196-T03-IM6-20,<br>EVT-P-0013196-T04-IM6-25                                                                          |
| 15         | 88512000  | 88513000 | 3        | EVT-P-0027006-T01-IM6-39,<br>EVT-P-0027006-T03-IM6-40,<br>EVT-P-0027006-T04-IM6-41                                                                          |

| chromosome | bin_start | bin_end  | n_events | event_ids                                                                          |
|------------|-----------|----------|----------|------------------------------------------------------------------------------------|
| 15         | 88600000  | 88601000 | 3        | EVT-P-0007772-T01-IM5-53,<br>EVT-P-0007772-T02-IM5-50,<br>EVT-P-0007772-T03-IM5-49 |
| 15         | 88548000  | 88549000 | 2        | EVT-P-0014872-T02-IM6-23,<br>EVT-P-0014872-T03-IM6-24                              |
| 15         | 88558000  | 88559000 | 2        | EVT-P-0013664-T02-IM5-1,<br>EVT-P-0013664-T04-IM6-16                               |
| 15         | 88651000  | 88652000 | 2        | EVT-P-0026390-T01-IM6-37,<br>EVT-P-0035552-T01-IM6-51                              |
| 15         | 88492000  | 88493000 | 1        | EVT-P-0058408-T01-IM6-3                                                            |
| 15         | 88497000  | 88498000 | 1        | EVT-P-0040355-T01-IM6-7                                                            |
| 15         | 88498000  | 88499000 | 1        | EVT-P-0006480-T01-IM5-54                                                           |
| 15         | 88513000  | 88514000 | 1        | EVT-P-0010668-T01-IM5-22                                                           |
| 15         | 88514000  | 88515000 | 1        | EVT-P-0018060-T01-IM6-31                                                           |
| 15         | 88522000  | 88523000 | 1        | EVT-P-0032148-T04-IM6-14                                                           |
| 15         | 88544000  | 88545000 | 1        | EVT-P-0025507-T01-IM6-34                                                           |
| 15         | 88550000  | 88551000 | 1        | EVT-P-0021974-T01-IM6-32                                                           |
| 15         | 88554000  | 88555000 | 1        | EVT-P-0061265-T01-IM7-2                                                            |
| 15         | 88556000  | 88557000 | 1        | EVT-P-0042503-T01-IM6-6                                                            |
| 15         | 88564000  | 88565000 | 1        | EVT-P-0064964-T01-IM7-15                                                           |
| 15         | 88576000  | 88577000 | 1        | EVT-P-0010556-T01-IM5-70                                                           |
| 15         | 88581000  | 88582000 | 1        | EVT-P-0010562-T01-IM5-36                                                           |
| 15         | 88588000  | 88589000 | 1        | EVT-P-0010264-T01-IM5-46                                                           |
| 15         | 88592000  | 88593000 | 1        | EVT-P-0007911-T02-IM5-47                                                           |
| 15         | 88594000  | 88595000 | 1        | EVT-P-0037610-T01-IM6-8                                                            |
| 15         | 88611000  | 88612000 | 1        | EVT-P-0026286-T01-IM6-44                                                           |
| 15         | 88613000  | 88614000 | 1        | EVT-P-0025987-T01-IM6-35                                                           |
| 15         | 88624000  | 88625000 | 1        | EVT-P-0056156-T02-IM7-4                                                            |
| 15         | 88632000  | 88633000 | 1        | EVT-P-0004737-T01-IM5-57                                                           |
| 15         | 88657000  | 88658000 | 1        | EVT-P-0026696-T01-IM6-38                                                           |
| 15         | 88662000  | 88663000 | 1        | EVT-P-0023257-T01-IM6-58                                                           |
| 15         | 88683000  | 88684000 | 1        | EVT-P-0058230-T02-IM7-19                                                           |

#### *exact\_position\_recurrence*

| chromosome | position | n_events | event_ids                                                                                                                                             |
|------------|----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15         | 88506571 | 6        | EVT-P-0002978-T02-IM5-43, EVT-P-0002978-T03-IM5-45,<br>EVT-P-0002978-T04-IM5-55, EVT-P-0002978-T05-IM6-56,<br>EVT-P-0002978-T06-IM6-21, ... (6 total) |

| chromosome | position | n_events | event_ids                                                                                              |
|------------|----------|----------|--------------------------------------------------------------------------------------------------------|
| 15         | 88488820 | 4        | EVT-P-0016507-T01-IM6-26, EVT-P-0016507-T02-IM6-27, EVT-P-0016507-T03-IM6-28, EVT-P-0016507-T04-IM6-29 |
| 15         | 88516534 | 4        | EVT-P-0033921-T02-IM6-13, EVT-P-0033921-T03-IM6-48, EVT-P-0033921-T04-IM6-12, EVT-P-0033921-T05-IM7-11 |
| 15         | 88501357 | 3        | EVT-P-0013196-T02-IM6-33, EVT-P-0013196-T03-IM6-20, EVT-P-0013196-T04-IM6-25                           |
| 15         | 88505787 | 3        | EVT-P-0035767-T01-IM6-52, EVT-P-0035767-T04-IM6-10, EVT-P-0035767-T06-IM7-9                            |
| 15         | 88512343 | 3        | EVT-P-0027006-T01-IM6-39, EVT-P-0027006-T03-IM6-40, EVT-P-0027006-T04-IM6-41                           |
| 15         | 88600844 | 3        | EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49                           |
| 15         | 88548029 | 2        | EVT-P-0014872-T02-IM6-23, EVT-P-0014872-T03-IM6-24                                                     |
| 15         | 88558191 | 2        | EVT-P-0013664-T02-IM5-1, EVT-P-0013664-T04-IM6-16                                                      |

### *protein\_position\_genomic\_spread*

| protein_position_aa | n_events | n_distinct_genomic_positions | genomic_span_bp | spans_multiple_chromosomes |
|---------------------|----------|------------------------------|-----------------|----------------------------|
| 529                 | 40       | 21                           | 75862           | False                      |
| 466                 | 15       | 13                           | 80532           | False                      |

## window\_detection tables

### *window\_scan*

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio          | p_value             | neg_log10_p_value  | event_ids_inside                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|---------------------|---------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 266      | 466    | 200      | 15              | 42               | 14                | 41                 | 0.34146341463414637 | 0.46052631578947384 | 0.3367455479305156 | EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (15 total) |
| 272      | 472    | 200      | 16              | 41               | 15                | 40                 | 0.375               | 0.486215538847118   | 0.313171165756522  | EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (16 total) |
| 366      | 466    | 100      | 15              | 42               | 14                | 41                 | 0.34146341463414637 | 0.46052631578947384 | 0.3367455479305156 | EVT-P-0004737-T01-IM5-57, EVT-P-0007772-T01-IM5-53, EVT-P-0007772-T02-IM5-50, EVT-P-0007772-T03-IM5-49, EVT-P-0007911-T02-IM5-47, ... (15 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio          | p_value             | neg_log10_p_value  | event_ids_inside                                                                                                                                                |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|---------------------|---------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 372      | 472    | 100      | 16              | 41               | 15                | 40                 | 0.375               | 0.486215538847118   | 0.313171165756522  | EVT-P-0004737-T01-IM5-57,<br>EVT-P-0007772-T01-IM5-53,<br>EVT-P-0007772-T02-IM5-50,<br>EVT-P-0007772-T03-IM5-49,<br>EVT-P-0007911-T02-IM5-47,<br>... (16 total) |
| 416      | 466    | 50       | 15              | 42               | 14                | 41                 | 0.34146341463414637 | 0.46052631578947384 | 0.3367455479305156 | EVT-P-0004737-T01-IM5-57,<br>EVT-P-0007772-T01-IM5-53,<br>EVT-P-0007772-T02-IM5-50,<br>EVT-P-0007772-T03-IM5-49,<br>EVT-P-0007911-T02-IM5-47,<br>... (15 total) |
| 422      | 472    | 50       | 16              | 41               | 15                | 40                 | 0.375               | 0.486215538847118   | 0.313171165756522  | EVT-P-0004737-T01-IM5-57,<br>EVT-P-0007772-T01-IM5-53,<br>EVT-P-0007772-T02-IM5-50,<br>EVT-P-0007772-T03-IM5-49,<br>EVT-P-0007911-T02-IM5-47,<br>... (16 total) |
| 441      | 466    | 25       | 15              | 42               | 14                | 41                 | 0.34146341463414637 | 0.46052631578947384 | 0.3367455479305156 | EVT-P-0004737-T01-IM5-57,<br>EVT-P-0007772-T01-IM5-53,<br>EVT-P-0007772-T02-IM5-50,<br>EVT-P-0007772-T03-IM5-49,<br>EVT-P-0007911-T02-IM5-47,<br>... (15 total) |
| 447      | 472    | 25       | 16              | 41               | 15                | 40                 | 0.375               | 0.486215538847118   | 0.313171165756522  | EVT-P-0004737-T01-IM5-57,<br>EVT-P-0007772-T01-IM5-53,<br>EVT-P-0007772-T02-IM5-50,<br>EVT-P-0007772-T03-IM5-49,<br>EVT-P-0007911-T02-IM5-47,<br>... (16 total) |
| 466      | 491    | 25       | 16              | 41               | 15                | 40                 | 0.375               | 0.486215538847118   | 0.313171165756522  | EVT-P-0004737-T01-IM5-57,<br>EVT-P-0007772-T01-IM5-53,<br>EVT-P-0007772-T02-IM5-50,<br>EVT-P-0007772-T03-IM5-49,<br>EVT-P-0007911-T02-IM5-47,<br>... (16 total) |
| 466      | 516    | 50       | 16              | 41               | 15                | 40                 | 0.375               | 0.486215538847118   | 0.313171165756522  | EVT-P-0004737-T01-IM5-57,<br>EVT-P-0007772-T01-IM5-53,<br>EVT-P-0007772-T02-IM5-50,<br>EVT-P-0007772-T03-IM5-49,<br>EVT-P-0007911-T02-IM5-47,<br>... (16 total) |
| 472      | 572    | 100      | 41              | 16               | 41                | 14                 | None                | 0.07518796992481204 | 1.1238516409670858 | EVT-P-0002671-T01-IM3-30,<br>EVT-P-0002978-T02-IM5-43,<br>EVT-P-0002978-T03-IM5-45,<br>EVT-P-0002978-T04-IM5-55,<br>EVT-P-0002978-T05-IM6-56,<br>... (41 total) |
| 472      | 672    | 200      | 41              | 16               | 41                | 14                 | None                | 0.07518796992481204 | 1.1238516409670858 | EVT-P-0002671-T01-IM3-30,<br>EVT-P-0002978-T02-IM5-43,<br>EVT-P-0002978-T03-IM5-45,<br>EVT-P-0002978-T04-IM5-55,<br>EVT-P-0002978-T05-IM6-56,<br>... (41 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio | p_value             | neg_log10_p_value | event_ids_inside                                                                                                                                                |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|------------|---------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 479      | 529    | 50       | 40              | 17               | 40                | 15                 | None       | 0.08521303258145364 | 1.069493978644493 | EVT-P-0002671-T01-IM3-30,<br>EVT-P-0002978-T02-IM5-43,<br>EVT-P-0002978-T03-IM5-45,<br>EVT-P-0002978-T04-IM5-55,<br>EVT-P-0002978-T05-IM6-56,<br>... (40 total) |
| 504      | 529    | 25       | 40              | 17               | 40                | 15                 | None       | 0.08521303258145364 | 1.069493978644493 | EVT-P-0002671-T01-IM3-30,<br>EVT-P-0002978-T02-IM5-43,<br>EVT-P-0002978-T03-IM5-45,<br>EVT-P-0002978-T04-IM5-55,<br>EVT-P-0002978-T05-IM6-56,<br>... (40 total) |
| 529      | 554    | 25       | 40              | 17               | 40                | 15                 | None       | 0.08521303258145364 | 1.069493978644493 | EVT-P-0002671-T01-IM3-30,<br>EVT-P-0002978-T02-IM5-43,<br>EVT-P-0002978-T03-IM5-45,<br>EVT-P-0002978-T04-IM5-55,<br>EVT-P-0002978-T05-IM6-56,<br>... (40 total) |
| 529      | 579    | 50       | 40              | 17               | 40                | 15                 | None       | 0.08521303258145364 | 1.069493978644493 | EVT-P-0002671-T01-IM3-30,<br>EVT-P-0002978-T02-IM5-43,<br>EVT-P-0002978-T03-IM5-45,<br>EVT-P-0002978-T04-IM5-55,<br>EVT-P-0002978-T05-IM6-56,<br>... (40 total) |
| 529      | 629    | 100      | 40              | 17               | 40                | 15                 | None       | 0.08521303258145364 | 1.069493978644493 | EVT-P-0002671-T01-IM3-30,<br>EVT-P-0002978-T02-IM5-43,<br>EVT-P-0002978-T03-IM5-45,<br>EVT-P-0002978-T04-IM5-55,<br>EVT-P-0002978-T05-IM6-56,<br>... (40 total) |
| 529      | 729    | 200      | 40              | 17               | 40                | 15                 | None       | 0.08521303258145364 | 1.069493978644493 | EVT-P-0002671-T01-IM3-30,<br>EVT-P-0002978-T02-IM5-43,<br>EVT-P-0002978-T03-IM5-45,<br>EVT-P-0002978-T04-IM5-55,<br>EVT-P-0002978-T05-IM6-56,<br>... (40 total) |

### top\_windows

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio | p_value             | neg_log10_p_value  | event_ids_inside                                                                                                                                                |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|------------|---------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 472      | 572    | 100      | 41              | 16               | 41                | 14                 | None       | 0.07518796992481204 | 1.1238516409670858 | EVT-P-0002671-T01-IM3-30,<br>EVT-P-0002978-T02-IM5-43,<br>EVT-P-0002978-T03-IM5-45,<br>EVT-P-0002978-T04-IM5-55,<br>EVT-P-0002978-T05-IM6-56,<br>... (41 total) |
| 504      | 529    | 25       | 40              | 17               | 40                | 15                 | None       | 0.08521303258145364 | 1.069493978644493  | EVT-P-0002671-T01-IM3-30,<br>EVT-P-0002978-T02-IM5-43,<br>EVT-P-0002978-T03-IM5-45,<br>EVT-P-0002978-T04-IM5-55,<br>EVT-P-0002978-T05-IM6-56,<br>... (40 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio          | p_value             | neg_log10_p_value  | event_ids_inside                                                                                                                                                |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|---------------------|---------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 441      | 466    | 25       | 15              | 42               | 14                | 41                 | 0.34146341463414637 | 0.46052631578947384 | 0.3367455479305156 | EVT-P-0004737-T01-IM5-57,<br>EVT-P-0007772-T01-IM5-53,<br>EVT-P-0007772-T02-IM5-50,<br>EVT-P-0007772-T03-IM5-49,<br>EVT-P-0007911-T02-IM5-47,<br>... (15 total) |
| 447      | 472    | 25       | 16              | 41               | 15                | 40                 | 0.375               | 0.486215538847118   | 0.313171165756522  | EVT-P-0004737-T01-IM5-57,<br>EVT-P-0007772-T01-IM5-53,<br>EVT-P-0007772-T02-IM5-50,<br>EVT-P-0007772-T03-IM5-49,<br>EVT-P-0007911-T02-IM5-47,<br>... (16 total) |

## Per-event results (results.tsv)

| event_id                | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                           | source_site_effect_on_frame |
|-------------------------|-------------------|------------|-----------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| EVT-P-0013664-T02-IM5-1 | P-0013664-T02-IM5 |            | Soft Tissue Sarcoma   | FIBS          | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88558191           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q26.1)(chr12:g.12032077::chr15:g.88558191)   Protein fusion: in frame (ETV6-NTRK3) | NA                          |
| EVT-P-0061265-T01-IM7-2 | P-0061265-T01-IM7 |            | Salivary Gland Cancer | HNMC          | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88554122           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12033175::chr15:g.88554122)   Protein Fusion: in frame {ETV6:NTRK3}                                                | NA                          |

| event_id                | sample_id         | patient_id | tumor_type        | oncotre_e_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                            | source_site2_effect_on_frame |
|-------------------------|-------------------|------------|-------------------|----------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0058408-T01-IM6-3 | P-0058408-T01-IM6 |            | Colorectal Cancer | COAD           | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88492156           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12035760::chr15:g.88492156)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |
| EVT-P-0056156-T02-IM7-4 | P-0056156-T02-IM7 |            | Thyroid Cancer    | THPA           | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88624274           | 13              | 466                         | True                   | 466                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12011957::chr15:g.88624274)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                | sample_id         | patient_id | tumor_type                | oncotre_e_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                            | source_site2_effect_on_frame |
|-------------------------|-------------------|------------|---------------------------|----------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0055703-T01-IM6-5 | P-0055703-T01-IM6 |            | Cancer of Unknown Primary | CUP            | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88505980           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12024785::chr15:g.88505980)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |
| EVT-P-0042503-T01-IM6-6 | P-0042503-T01-IM6 |            | Colorectal Cancer         | COAD           | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88556067           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12032802::chr15:g.88556067)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                | sample_id         | patient_id | tumor_type        | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                   | source_site2_effect_on_frame |
|-------------------------|-------------------|------------|-------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0040355-T01-IM6-7 | P-0040355-T01-IM6 |            | Colorectal Cancer | COAD          | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88497191           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12036111::chr15:g.88497191)   Protein Fusion: in frame {ETV6:NTRK3}        | NA                           |
| EVT-P-0037610-T01-IM6-8 | P-0037610-T01-IM6 |            | Thyroid Cancer    | THPA          | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88594532           | 13              | 466                         | True                   | 466                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) rearrangement: t(12;15)(p13.2;q25.3)(chr12:g.12014663::chr15:g.88594532)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                 | sample_id          | patient_id | tumor_type        | oncotre_e_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                            | source_site2_effect_on_frame |
|--------------------------|--------------------|------------|-------------------|----------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0035767-T06-IM7-9  | P-0035767-T06-I M7 |            | Pancreatic Cancer | PAAC           | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88505787           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |
| EVT-P-0035767-T04-IM6-10 | P-0035767-T04-I M6 |            | Pancreatic Cancer | PAAD           | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88505787           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type    | oncotre_e_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                            | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|---------------|----------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0033921-T05-IM7-11 | P-0033921-T05-IM7 |            | Breast Cancer | JSCB           | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88516534           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |
| EVT-P-0033921-T04-IM6-12 | P-0033921-T04-IM6 |            | Breast Cancer | JSCB           | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88516534           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                            | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-----------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0033921-T02-IM6-13 | P-0033921-T02-IM6 |            | Breast Cancer         | JSCB          | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88516534           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |
| EVT-P-0032148-T04-IM6-14 | P-0032148-T04-IM6 |            | Salivary Gland Cancer | HNMA5C        | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88522437           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12037217::chr15:g.88522437)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type          | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|---------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0064964-T01-IM7-15 | P-0064964-T01-IM7 |            | Breast Cancer       | JSCB          | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88564682           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12032310::chr15:g.88564682)   Protein Fusion: in frame {ETV6:NTRK3}                                                      | NA                           |
| EVT-P-0013664-T04-IM6-16 | P-0013664-T04-IM6 |            | Soft Tissue Sarcoma | FIBS          | ETV6-NTRK3  | ETV6         | in-frame     | three_prime | 88558191           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032077::chr15:g.88558191)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type        | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail                                                                                                                                                                             | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                                               | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------|---------------------|-----------------------|-----------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0058230-T02-IM7-19 | P-0058230-T02-IM7 |            | Pancreatic Cancer | PAAD          | NTRK3-DOT1L | DOT1L        | out-of-frame | five_prime  | 88683244           | 5               | 155                         | True                   | 154                                          | True                               | hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | lost          | 0.0                      | False               |                       | Protein kinase domain |                   | NTRK3 (NM_001012338) - DOT1L (NM_032482) rearrangement: t(15;19)(q25.3;p13.3)(chr15:g.88683244::chr19:g.2227604)   Protein Fusion: out of frame {NTRK3:DOT1L}                        | NA                           |
| EVT-P-0013196-T03-IM6-20 | P-0013196-T03-IM6 |            | Breast Cancer     | BNNOS         | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88501357           | 14              | 529                         | True                   | 529                                          | True                               | exact match                                                                                                                                                                                                    | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357)   Protein fusion: in frame (ETV6-NTRK3) | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-----------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0002978-T06-IM6-21 | P-0002978-T06-IM6 |            | Salivary Gland Cancer | HNMAC         | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88506571           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571)   Protein fusion: in frame (ETV6-NTRK3) | NA                           |
| EVT-P-0010668-T01-IM5-22 | P-0010668-T01-IM5 |            | Salivary Gland Cancer | HNMAC         | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88513282           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029109::chr15:g.88513282)   Protein fusion: in frame (ETV6-NTRK3)             | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type          | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                     | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|---------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0014872-T02-IM6-23 | P-0014872-T02-IM6 |            | Soft Tissue Sarcoma | IMT           | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88548029           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946::chr15:g.88548029)   Protein fusion: in frame (ETV6-NTRK3) | NA                           |
| EVT-P-0014872-T03-IM6-24 | P-0014872-T03-IM6 |            | Soft Tissue Sarcoma | IMT           | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88548029           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946::chr15:g.88548029)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-----------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0013196-T04-IM6-25 | P-0013196-T04-IM6 |            | Breast Cancer         | BRCNO S       | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88501357           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357)   Protein Fusion: in frame {ETV6:NTRK3}                   | NA                           |
| EVT-P-0016507-T01-IM6-26 | P-0016507-T01-IM6 |            | Salivary Gland Cancer | HNMAS C       | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88488820           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                                      | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-----------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0016507-T02-IM6-27 | P-0016507-T02-IM6 |            | Salivary Gland Cancer | HNMAC         | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88488820           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-14 fused with NTRK3 exons 6-20 including the kinase domain): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                         | source_site2_effect_on_frame |
|-------------------------|-------------------|------------|-----------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-016507-T03-IM6-28 | P-0016507-T03-IM6 |            | Salivary Gland Cancer | HNMC          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88488820           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20: t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |
| EVT-P-016507-T04-IM6-29 | P-0016507-T04-IM6 |            | Salivary Gland Cancer | HNMC          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88488820           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused to NTRK3 exons 15-20 ): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type            | oncotre_e_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-----------------------|----------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|-------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0002671-T01-IM3-30 | P-0002671-T01-IM3 |            | Colorectal Cancer     | COAD           | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88505784           | 14              | 529                         | True                   | 529                                           | True                                | exact match                         | retained      | 1.0                      | False               | Protein kinase domain |              |                   | NA   Protein fusion: in frame (ETV6-NTRK3)                                                                                                                                                             | NA                           |
| EVT-P-0018060-T01-IM6-31 | P-0018060-T01-IM6 |            | Salivary Gland Cancer | HNMSC          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88514531           | 14              | 529                         | True                   | 529                                           | True                                | exact match                         | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037::chr15:g.88514531)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                        | source_site2_effect_on_frame |
|-------------------------|-------------------|------------|-----------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-021974-T01-IM6-32 | P-0021974-T01-IM6 |            | Salivary Gland Cancer | SAAD          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88550500           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028417::chr15:g.88550500)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |
| EVT-P-013196-T02-IM6-33 | P-0013196-T02-IM6 |            | Breast Cancer         | BRCNOS        | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88501357           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357)   Protein fusion: in frame (ETV6-NTRK3)          | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type                | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                          | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|---------------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002507-T01-IM6-34  | P-0025507-T01-IM6 |            | Breast Cancer             | IDC           | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88544573           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12024078::chr15:g.88544573)   Protein Fusion: in frame {ETV6:NTRK3}      | NA                           |
| EVT-P-0025987-T01-IM6-35 | P-0025987-T01-IM6 |            | Miscellaneous Brain Tumor | PRNET         | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88613114           | 13              | 466                         | True                   | 466                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12028942::chr15:g.88613114)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                          | source_site2_effect_on_frame |
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| EVT-P-0010562-T01-IM5-36 | P-0010562-T01-IM5 |            | Salivary Gland Cancer | SDCA          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88581658           | 13              | 466                         | True                   | 466                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 14-20): t(12;15)(p13.2;q25.1)(chr12:g.12028903::chr15:g.88581658)   Protein fusion: in frame (ETV6-NTRK3)       | NA                           |
| EVT-P-0026390-T01-IM6-37 | P-0026390-T01-IM6 |            | Thyroid Cancer        | THPA          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88651131           | 13              | 466                         | True                   | 466                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12016863::chr15:g.88651131)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type          | oncogene_code | fusion_name | partner_gene | frameshift | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                          | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|---------------------|---------------|-------------|--------------|------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0026696-T01-IM6-38 | P-0026696-T01-IM6 |            | Thyroid Cancer      | THPA          | NTRK3-ETV6  | ETV6         | in-frame   | three_prime | 88657925           | 13              | 466                         | True                   | 466                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12007967::chr15:g.88657925)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |
| EVT-P-0027006-T01-IM6-39 | P-0027006-T01-IM6 |            | Soft Tissue Sarcoma | SARCN OS      | NTRK3-ETV6  | ETV6         | in-frame   | three_prime | 88512343           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352::chr15:g.88512343)   Protein Fusion: in frame {ETV6:NTRK3}       | NA                           |

| event_id                | sample_id          | patient_id | tumor_type          | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                  | source_site2_effect_on_frame |
|-------------------------|--------------------|------------|---------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-027006-T03-IM6-40 | P-0027006-T03-I M6 |            | Soft Tissue Sarcoma | FIBS          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88512343           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037::chr15:g.88514531)   Protein Fusion: in frame {ETV6:N TRK3} | NA                           |
| EVT-P-027006-T04-IM6-41 | P-0027006-T04-I M6 |            | Soft Tissue Sarcoma | FIBS          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88512343           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352::chr15:g.88512343)   Protein Fusion: in frame {ETV6:N TRK3}        | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_frame |
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| EVT-P-0002978-T07-IM6-42 | P-0002978-T07-IM6 |            | Salivary Gland Cancer | HNMC          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88506571           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |
| EVT-P-0002978-T02-IM5-43 | P-0002978-T02-IM5 |            | Salivary Gland Cancer | HNMC          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88506571           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | NA   Protein fusion: in frame (ETV6-NTRK3)                                                                                                                                                             | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                               | source_site2_effect_on_frame |
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| EVT-P-0026286-T01-IM6-44 | P-0026286-T01-IM6 |            | Salivary Gland Cancer | SACA          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88611476           | 13              | 466                         | True                   | 466                                           | True                                | exact match                         | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12030395::chr15:g.88611476)   Protein Fusion: in frame {ETV6:NTRK3}      | NA                           |
| EVT-P-0002978-T03-IM5-45 | P-0002978-T03-IM5 |            | Salivary Gland Cancer | HNMA5C        | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88506571           | 14              | 529                         | True                   | 529                                           | True                                | exact match                         | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 5 fused to NTRK3 exons 15 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571)   Protein fusion: in frame (ETV6-NTRK3) | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type        | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                   | source_site_effect_on_frame |
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| EVT-P-0010264-T01-IM5-46 | P-0010264-T01-IM5 |            | Pancreatic Cancer | PAAD          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88588255           | 13              | 466                         | True                   | 466                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused in frame with NTRK3 exons 13-20): t(12;15)(p13.2;q25.3)(chr12:g.12014962::chr15:g.88588255)   Protein fusion: in frame (ETV6-NTRK3) | NA                          |
| EVT-P-0007911-T02-IM5-47 | P-0007911-T02-IM5 |            | Thyroid Cancer    | THPD          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88592430           | 13              | 466                         | True                   | 466                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 4 fused with NTRK3 exons 13-20): t(12;15)(p13.2;q25.3)(chr12:g.12007142::chr15:g.88592430)   Protein fusion: in frame (ETV6-NTRK3)       | NA                          |

| event_id                 | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                      | source_site2_effect_on_frame |
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| EVT-P-0033921-T03-IM6-48 | P-0033921-T03-IM6 |            | Breast Cancer         | BRCNOS        | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88516534           | 14              | 529                         | True                   | 529                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534)   Protein Fusion: in frame {ETV6:NTRK3}                                                           | NA                           |
| EVT-P-007772-T03-IM5-49  | P-007772-T03-IM5  |            | Salivary Gland Cancer | HNMASC        | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88600844           | 13              | 466                         | True                   | 466                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844)   Protein fusion: in frame (ETV6-NTRK3) | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail                                                                                                                                                                            | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                                                                      | source_site_effect_on_frame |
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| EVT-P-0007772-T02-IM5-50 | P-0007772-T02-IM5 |            | Salivary Gland Cancer | HNMAC         | NTRK3-ETV6  | ETV6         | in-frame     | five_prime  | 8860084            | 12              | 466                         | True                   | 465                                           | True                                | hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | lost          | 0.0                      | False               |                       | Protein kinase domain |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844)   Protein fusion: in frame (NTRK3-ETV6) | NA                          |
| EVT-P-0035552-T01-IM6-51 | P-0035552-T01-IM6 |            | Pancreatic Cancer     | PAAD          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 8865125            | 13              | 466                         | True                   | 466                                           | True                                | exact match                                                                                                                                                                                                    | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12020963::chr15:g.88651251)   Protein Fusion: in frame {ETV6:NTRK3}             | NA                          |

| event_id                 | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                      | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-----------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|-------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0035767-T01-IM6-52 | P-0035767-T01-IM6 |            | Pancreatic Cancer     | PANET         | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88505787           | 14              | 529                         | True                   | 529                                           | True                                | exact match                         | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787)   Protein Fusion: in frame {ETV6:NTRK3}                                                           | NA                           |
| EVT-P-007772-T01-IM5-53  | P-007772-T01-IM5  |            | Salivary Gland Cancer | HNMC          | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88600844           | 13              | 466                         | True                   | 466                                           | True                                | exact match                         | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844)   Protein fusion: in frame (ETV6-NTRK3) | NA                           |

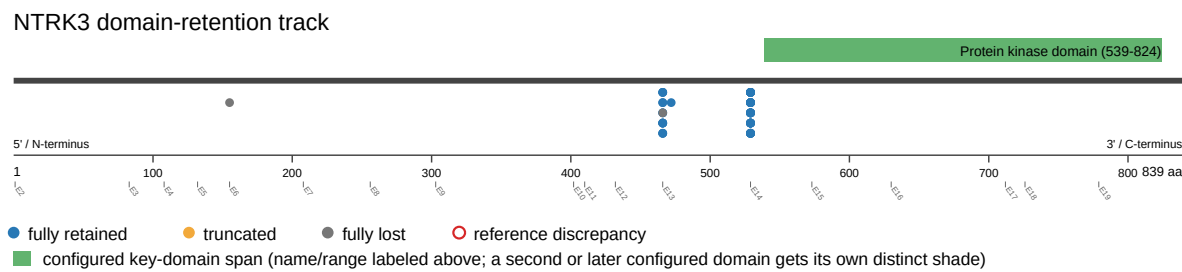
| event_id                 | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                      | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-----------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|-------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0006480-T01-IM5-54 | P-0006480-T01-IM5 |            | Salivary Gland Cancer | HNMAC         | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88498145           | 14              | 529                         | True                   | 529                                           | True                                | exact match                         | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029548::chr15:g.88498145)   Protein fusion: in frame (ETV6-NTRK3) | NA                           |
| EVT-P-0002978-T04-IM5-55 | P-0002978-T04-IM5 |            | Salivary Gland Cancer | HNMAC         | NTRK3-ETV6  | ETV6         | in-frame     | three_prime | 88506571           | 14              | 529                         | True                   | 529                                           | True                                | exact match                         | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571)   Protein fusion: in frame (ETV6-NTRK3) | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name | partner_gene | frameshift | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intro_nic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-----------------------|---------------|-------------|--------------|------------|-------------|--------------------|-----------------|-----------------------------|-------------------------|-----------------------------------------------|-------------------------------------|-------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0002978-T05-IM6-56 | P-0002978-T05-IM6 |            | Salivary Gland Cancer | HNMAC         | NTRK3-ETV6  | ETV6         | in-frame   | three_prime | 88506571           | 14              | 529                         | True                    | 529                                           | True                                | exact match                         | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571)   Protein fusion: in frame (ETV6-NTRK3) | NA                           |
| EVT-P-0004737-T01-IM5-57 | P-0004737-T01-IM5 |            | Pancreatic Cancer     | PAAD          | NTRK3-ETV6  | ETV6         | in-frame   | three_prime | 88632786           | 13              | 466                         | True                    | 466                                           | True                                | exact match                         | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 13 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12006809::chr15:g.88632786)   Protein fusion: in frame (ETV6-NTRK3)   | NA                           |

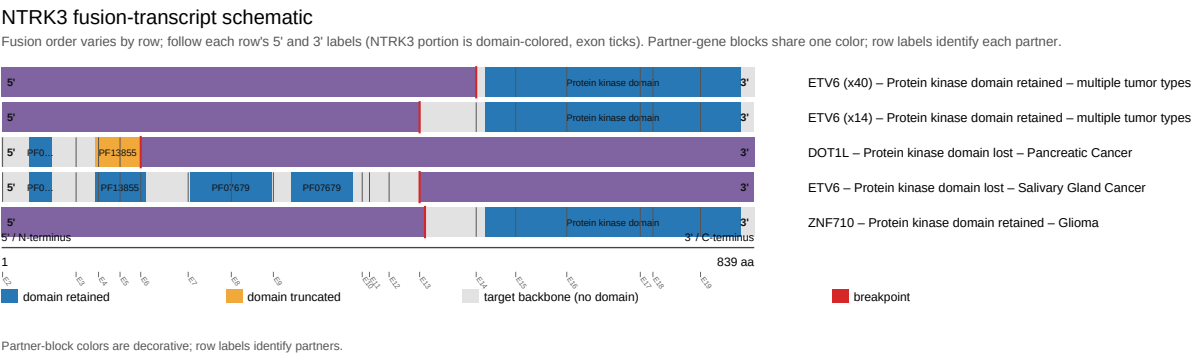
| event_id                | sample_id         | patient_id | tumor_type            | oncogene_code | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                               | source_site2_effect_on_frame |
|-------------------------|-------------------|------------|-----------------------|---------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-023257-T01-IM6-58 | P-0023257-T01-IM6 |            | Salivary Gland Cancer | HNMAC         | NTRK3-ETV6   | ETV6         | in-frame     | three_prime | 88662190           | 13              | 466                         | True                   | 466                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 14 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12014998::chr15:g.88662190)   Protein Fusion: in frame {ETV6:NTRK3} | NA                           |
| EVT-P-010556-T01-IM5-70 | P-0010556-T01-IM5 |            | Glioma                | GB            | NTRK3-ZNF710 | ZNF710       | unknown      | three_prime | 88576258           | 13              | 472                         | False                  | 472                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | NTRK3 (NM_001012338) - ZNF710 (NM_198526) rearrangement: c.-28-10227:ZNF710_c.1415:NTRK3inv   Protein fusion: mid-exon (ZNF710-NTRK3)                                                                | NA                           |

# Figures

## domain retention outliers



## fusion schematic



## intragenic deletion schematic

